

Code:

#prog1

print("Hello World")

print()

#prog2

usertext = input("What is your name? ")

print("Hello", usertext)

print()

#prog3

num1 = input('Enter first number: ')

num2 = input('Enter second number: ')

sum = float(num1) + float(num2)

print('The sum of {0} and {1} is {2}'.format(num1, num2, sum))

print()

#prog4

num1 = input('Enter first number: ')

num2 = input('Enter second number: ')

average =(int(num1) + int(num2))

print('average:{0} '.format(average))

print()

#prog5

visagrade = input('enter your visa grade : ')

finalgrade = input('enter your final grade : ')

average =(float(visagrade)*0.3)+(float(finalgrade)*0.7)

print("average :{0} ".format(average))

print()

#prog6

firstexam = input('your first exam : ')

secondexam = input('your second exam : ')

thirddexam = input('your third exam : ')

average =(float(firstexam)+float(secondexam)+float(thirddexam))/3

print("average :{0} ".format(average))

print()

#prog7

average = input('enter average : ')

if(int(average)>=50):

```
    print("Passed")
```

```
else:
```

```
    print("Failed")
```

```
print()
```

```
#prog8
```

```
num = int(input("Enter a number: "))
```

```
if (num % 2) == 0:
```

```
    print("{0} is Even".format(num))
```

```
else:
```

```
    print("{0} is Odd".format(num))
```

```
print()
```

```
#prog9
```

```
num = float(input("Enter a number: "))
```

```
if num > 0:
```

```
    print("Positive number")
```

```
elif num == 0:
```

```
    print("Zero")
```

```
else:
```

```
    print("Negative number")
```

```
print()
```

```
#prog10
```

```
print("body mass index calculation program")
```

```
height = float(input("enter height (m):"))
```

```
weight = int(input("enter weight (kg):"))
```

```
index = weight/(height*height)
```

```
if index <=18:
```

```
    print("\n underweight BMİ:{0}".format(index))
```

```
elif index > 18 and index <=25 :
```

```
    print("\n overweight BMİ:{0}".format(index))
```

```
elif index > 25 and index <=30:
```

```
    print("\n obese BMİ:{0}".format(index))
```

```
elif index > 30:
```

```
    print("\n severely obese BMİ:{0}".format(index))
```

```
print()
```

```
#prog11
```

```
age = input('enter age : ')
```

```
if(int(age)<18):
```

```
    print("Your Age Is Not Eligible To Get A Driver's License")
```

```
else:
```

```
    print("Your Age Is Eligible To Get Your License")
```

```
print()
```

```
#prog12
```

```
for i in range(1,101):
```

```
    print(i)
```

```
print()
```

```
#prog13
```

```
for i in range(1,101):
```

```
    if i%2==0:
```

```
        print(i)
```

```
print()
```

```
#prog14
```

```
for i in range(1,101):
```

```
    if i%2!=0:
```

```
        print(i)
```

```
print()
```

```
#prog15
```

```
for i in range(1,101):
```

```
    if i%3==0 or i%5==0:
```

```
        print(i)
```

```
print()
```

```
#prog16
```

```
num = input('enter number : ')
```

```
for i in range(1,int(num)+1):
```

```
    print(i)
```

```
print()
```

```
#prog17
```

```
short = input('Enter short side : ')
```

```
tall = input('Enter tall side : ')
```

```
area = int(short) * int(tall)
```

```
perimeter = 2 * (int(short) + int(tall))
```

```
print("area: {0}".format(area))
```

```
print("perimeter: {0}".format(perimeter))
```

```
print()
```

```
#prog18
```

```
word = 'mrhuseyin'
```

```
for char in word:
```

```
    print(char)
```

```
print()
```

```
#prog19
```

```
sumofnumbers = 0
```

```
num1 = input('first number: ')
```

```
num2 = input('second number: ')
```

```
start = int(num1)
```

```
end = int(num2)
```

```
low = min(start, end)
```

```
high = max(start, end)
```

```
for i in range(low + 1, high):
```

```
    sumofnumbers += i
```

```
print("Sum of numbers between {0} and {1} : {2}".format(num1, num2, sumofnumbers))
```

```
print()
```

```
#prog20
```

```
selection = input("Press (1) for Cinema, (2) for Theater : ")
```

```
student = input("Are you student(Y/N) : ")
```

```
price = 0
```

```
    #non-discounted fee calculation
```

```
if selection == '1':
```

```
    price = 10 #cinema
```

```
elif selection == '2':
```

```
    price = 5 #theatre
```

```
    #student discount
```

```
if student == 'Y' or student == 'y':
```

```
    price=price / 2 #%50
```

```
    print(" The fee you have to pay :{}".format(price))
```

```
print()
```

```
#prog21
```

```
num = int(input("Enter a number: "))
```

```
if num > 1:
```

```
    for i in range(2,num):
```

```
        if (num % i) == 0:
```

```
            print(num,"is not a prime number")
```

```
            print(i,"times",num//i,"is",num)
```

```
            break
```

```
    else:
```

```
        print(num,"is a prime number")
```

```
else:

    print(num,"is not a prime number")

print()


#prog22

NumList = []

Even_Sum = 0

Odd_Sum = 0


Number = int(input("Please enter the Total Number of List Elements: "))

for i in range(1, Number + 1):

    value = int(input("Please enter the Value of %d Element : " %i))

    NumList.append(value)


for j in range(Number):

    if(NumList[j] % 2 == 0):

        Even_Sum = Even_Sum + NumList[j]

    else:

        Odd_Sum = Odd_Sum + NumList[j]


print("\nThe Sum of Even Numbers in this List = ", Even_Sum)

print("The Sum of Odd Numbers in this List = ", Odd_Sum)

print()
```

```
#prog23

newsalary = 0

salary = input("enter new salary : ")


raise_rate = input("salary raise rate(%) : ")


newsalary = float(salary) + (float(salary) * float(raise_rate) / 100)


print("increased salary :", newsalary)

print()
```

```
#prog24

import math


def find_Diameter(radius):

    return 2 * radius


def find_Circumference(radius):

    return 2 * math.pi * radius


def find_Area(radius):

    return math.pi * radius * radius
```

```
r = float(input(' Please Enter the radius of a circle: '))
```

```
diameter = find_Diameter(r)
```

```
circumference = find_Circumference(r)
```

```
area = find_Area(r)
```

```
print("\n Diameter Of a Circle = %.2f" %diameter)
```

```
print(" Circumference Of a Circle = %.2f" %circumference)
```

```
print(" Area Of a Circle = %.2f" %area)
```

```
print()
```

```
#prog25
```

```
def areaRectangle(a, b):
```

```
    return (a * b)
```

```
def perimeterRectangle(a, b):
```

```
    return (2 * (a + b))
```

```
a = 5;
```

```
b = 6; print ("Area = ", areaRectangle(a, b))
```

```
print ("Perimeter = ", perimeterRectangle(a, b))
```

```
print()
```

```
#prog26
```

```
import random
```

```
import math
```

```
# Taking Inputs
```

```
lower = int(input("Enter Lower bound:- "))
```

```
upper = int(input("Enter Upper bound:- "))
```

```
# Generating random number between the lower and upper
```

```
x = random.randint(lower, upper)
```

```
# Calculate total chances cleanly using ceil
```

```
chances = math.ceil(math.log(upper - lower + 1, 2))
```

```
print(f"\n\tYou've only {chances} chances to guess the integer!\n")
```

```
# Initializing the number of guesses.
```

```
count = 0
```

```
# Loop runs while count is less than total allowed chances
```

```
while count < chances:
```

```
    # Taking guessing number as input
```

```
    guess = int(input("Guess a number:- "))
```

```
count += 1 # Increment after the guess is taken


# Condition testing

if x == guess:

    print("Congratulations you did it in ", count, " try")

    break

elif x > guess:

    print("You guessed too small!")

elif x < guess:

    print("You Guessed too high!")


# If the loop finishes and the last guess was wrong

if count >= chances and x != guess:

    print("\nThe number is %d" % x)

    print("\tBetter Luck Next time!")


print()


#prog27

import datetime

date=str(input('Enter the date(for example:09 02 2019):'))

day_name= ['Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday',
'Saturday','Sunday']

day = datetime.datetime.strptime(date, '%d %m %Y').weekday()

print(day_name[day])

print()


#prog28

def find_missing(lst):

    return [x for x in range(lst[0], lst[-1]+1)

if x not in lst]


# Driver code

lst = [1, 2, 4, 6, 7, 9, 10]

print(find_missing(lst))

print()


#prog29

char_list = ["a", "b" ,"c"]

string = "abcd"

matched_list = [characters in char_list for characters in string]

print(matched_list)

string_contains_chars = all(matched_list)

print(string_contains_chars)
```

```
print()

#prog30

total = 0

evenSums = 0

evenCount = 0


oddSums = 0

oddCount = 0


done = False


while not done:

    user_in = input("Give me an integer or type 'done' to be done: ")

    if user_in.lower() == "done":

        done = True

    else:

        num = int(user_in)

        total += num

        if num % 2 == 0:

            evenSums += num

            evenCount += 1

        else:

            oddSums += num

            oddCount += 1


evenAverage = evenSums / evenCount if evenCount > 0 else 0

oddAverage = oddSums / oddCount if oddCount > 0 else 0


print("\n--- Results ---")

print("Total Sum:", total)

print("Even Average: " + str(evenAverage))

print("Odd Average: " + str(oddAverage))
```

OUTPUT:

OUTPUT:

Hello World

What is your name? Shyra

Hello Shyra

Enter first number: 1

Enter second number: 2

The sum of 1 and 2 is 3.0

Enter first number: 1

Enter second number: 2

average:3

enter your visa grade: 87

enter your final grade: 89

average :88.39999999999999

your first exam: 99

your second exam: 87

your third exam: 25

average :70.33333333333333

enter average: 770

Passed

Enter a number: 1

1 is Odd

Enter a number: 2

Positive number

body mass index calculation program

enter height (m):176

enter weight (kg):45

underweight BMi:0.001452737603305785

enter age: 12

Your Age Is Not Eligible To Get A Driver's License

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99
100

enter number: 1

1

Enter short side: 12

Enter tall side: 14

area: 168

perimeter: 52

m

r

h

u

s

e

y

i

n

first number: 1

second number: 2

Sum of numbers between 1 and 2 : 0

Press (1) for Cinema, (2) for Theater: 1

Are you student(Y/N): y

The fee you have to pay :5.0

Enter a number: 12

12 is not a prime number

2 times 6 is 12

Please enter the Total Number of List Elements: 3

Please enter the Value of 1 Element: 10

Please enter the Value of 2 Element: 12

Please enter the Value of 3 Element: 2

The Sum of Even Numbers in this List = 24

The Sum of Odd Numbers in this List = 0

enter new salary: 1000

salary raise rate(%): 10

increased salary: 1100.0

Please Enter the radius of a circle: 12

Diameter Of a Circle = 24.00

Circumference Of a Circle = 75.40

Area Of a Circle = 452.39

Area = 30

Perimeter = 22

Enter Lower bound:- 12

Enter Upper bound:- 34

You've only 5 chances to guess the integer!

Guess a number:- 34

You Guessed too high!

Guess a number:- 20

You Guessed too high!

Guess a number:- 15

You Guessed too high!

Guess a number:- 10

You guessed too small!

Guess a number:- 13

You guessed too small!

The number is 14

Better Luck Next time!

Enter the date(for example:09 02 2019):06 07 2021

Tuesday

[3, 5, 8]

[True, True, True, False]

False

Give me an integer or type 'done' to be done: done

--- Results ---

Total Sum: 0

Even Average: 0

Odd Average: 0

PS C:\Users\SBH-CL5-WS01\shy>